GLYPH ANALYTICS

How Trimtab works

AI-assisted experiment design where AI proposes and critiques, deterministic code enforces the rules, and a human owner decides.

Human owner decides | AI proposes and attacks | Code enforces

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Most experiments fail before any data arrives

Wrong unit

Randomizing users when the treatment spills across a team, store or region.

Underpowered

Sample sizes guessed rather than computed from real baselines and variance.

Hidden assumptions

Interference, carryover or confounding that nobody wrote down.

Moving goalposts

Metrics or estimators chosen after the results come in.

Trimtab catches these at design time, and makes sure an AI assistant can’t introduce them.

Three roles, strictly separated

HUMAN OWNER

Decides

- Asks the question and locks known facts
- Selects the design
- Justifies every warning in writing
- Supplies baselines, variance, MDE
- Approves, on the record

AI (TWO SEPARATE ROLES)

Proposes and attacks

- Planner drafts 2–4 candidate designs
- Challenger, a separate call, critiques each one
- Advisory only: no approval, no numbers

DETERMINISTIC CODE

Enforces

- Scores designs PASS / WARN / FAIL
- Computes power and sample size
- Freezes, verifies assignment, gates launch
- Fails closed

From question to frozen design

1 · Human

Ask

Plain-language question; lock outcome, unit and arms

2 · AI

Propose

Planner drafts 2–4 candidate designs

3 · AI

Attack

Separate challenger critiques every candidate

4 · Code

Judge

PASS, WARN or FAIL; failed designs are blocked

5 · Human

Choose

Select; justify warnings; supply baseline, variance, MDE, traffic

6 · Code

Size

Power engine computes N; schema and methods validated

7 · Human + code

Freeze

Recorded approval; SHA-256 freeze makes it immutable

Every step’s output is checked before the next. A design can’t reach freeze without passing step 4, a justified warning list, and a recorded human approval.

Two AI calls, adversarial by design

Planner

- Turns a plain-language question into 2–4 scientifically distinct candidates
- Names the empirical inputs power will need, without supplying any

Challenger

- Hunts for broken identification, interference, carryover, unit mismatch, outcome leakage and weak instruments
- Marks each design KEEP, REVISE or REJECT, in the owner’s own vocabulary

What the challenger may not do

Choose a winner, declare a concern resolved, set gate consequences, or contradict a deterministic PASS. Its questions start a design dialogue: the owner answers, the planner patches, the challenger reviews again.

The human owner stays in control

The owner...	...and the system refuses to freeze unless
Locks known facts up front	Every candidate matches the locked outcome, unit and arms
Selects the design	The selected candidate is not BLOCKED; AI never selects
Justifies each warning	Every WARN carries a written rationale
Supplies empirical inputs	No __REQUIRED__ placeholders remain; power came from human or deterministic sources only
Confirms assumptions	Counterfactual and assumptions are explicitly confirmed
Approves	approved = true is set by the owner and recorded with name and time

The frozen design records the selected candidate, workflow hash, approver, approval time and every warning acknowledgment.

Enforced in code, not in prompts

Power firewall

AI output schemas have no numeric power fields. Only the deterministic engine computes N, power, MDE, allocation and timing.

No approval path

The schemas contain no approval field. “Mark this approved” inside a prompt does nothing; question text is inert data.

Field-level provenance

Every field in a candidate design records whether it came from the AI or from the owner’s locked facts.

No silent causal mistakes

Post-treatment variables can’t become adjustment covariates, and contracts can’t quietly override each other.

Six gates, all fail-closed

1 · Adjudication

Blocks invalid arms, unsafe randomization units, carryover, endogenous dose, or no analysis path

2 · Power firewall

Sample size, power and MDE come only from the deterministic engine and human inputs

3 · Freeze

Analysis, metric and validity contracts must agree; SHA-256 detects any later change

4 · Verified assignment

Rebuilt from the eligibility snapshot with an HMAC secret; corrupted data is rejected

5 · Launch gate

Packet integrity, power reconciliation, assignment, instrumentation probe and governance must all pass

6 · Frozen analysis

Prespecified estimator and metric family; SRM, balance and missingness flagged

After results: an AI critic reviews the evidence, bounded by a deterministic ship/no-ship policy on effect size, adjusted p-value, interval direction and rollout economics.

Tested against known truth

EFFECT INJECTED · DETECTED

+7.63 pp

Day-7 activation, 2,488 enrolled
95% interval +4.17 to +11.08 pp

NO EFFECT INJECTED · INCONCLUSIVE

+0.48

Day-7 task score, 1,085 enrolled
95% interval −0.71 to +1.67 points

Both runs reached complete, accepted frozen analysis. Also: adversarial validation suites, Monte Carlo estimator checks, and a semantic suite against live model output.

Thirteen design families, end to end

A/B and multi-arm tests	Blocked / stratified	Cluster randomized
Geographic randomized	Factorial (up to 16 cells)	Ordered dose-response
Switchback	Within-subject crossover	Stepped-wedge rollout
Encouragement / IV	Saturation and spillover	Nested multilevel
DiD, ITS, synthetic control		

“End to end” means candidate review through power, freeze, data mapping and confirmatory analysis. Designs with substantive assumptions are flagged “supported with limits” and routed to human review; anything without a full path can’t be selected.

AI widens the options.
Code enforces the rules.
People decide.

Case study and diagrams: github.com/klovelace

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